

SPEAKERS' BIOGRAPHIES



Rajesh Rajamani (Fellow, IEEE) received the B.Tech. degree from the Indian Institute of Technology Madras, Chennai, India, in 1989, and the M.S. and Ph.D. degrees from the University of California at Berkeley, Berkeley, CA, USA, in 1991 and 1993, respectively. He is currently the Benjamin Y. H. Li-TSI Endowed Professor of mechanical engineering and Associate Director of the Minnesota Robotics Institute, University of Minnesota, Minneapolis, MN, USA. He is the author of the popular book titled *Vehicle Dynamics and Control* (Springer). He has authored or coauthored over 180 journal papers and is a co-inventor on 14 granted patents. His active research interests include estimation, sensing, and control for smart mechanical systems. Dr. Rajamani has served as the Chair for the IEEE Technical Committee on Automotive Control and is currently a Senior Editor of the IEEE Transactions on Intelligent Transportation Systems. He is a fellow of ASME and was a recipient of the CAREER Award from the National Science Foundation, the Ralph Teetor Award from SAE, the Charles Stark Draper Award from ASME, the O. Hugo Schuck Award from the American Automatic Control Council, and a number of best paper awards from conferences and journals. Several inventions from his laboratory have been commercialized through startup ventures co-founded by industry executives. One of these companies, Innotronics, was recently recognized among the 35 Best University Start-Ups of 2016 by the U.S. National Council of Entrepreneurial Tech Transfer.



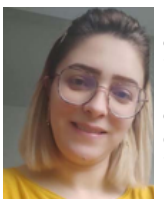
Vicenç Puig received his BS/MS degree in telecommunications engineering in 1993 and his PhD degree in Automatic Control, Robotics and Computer Vision in 1999, both from the Universitat Politècnica de Catalunya (UPC), Barcelona, Spain. He is a full professor at the Automatic Control Department of UPC and a researcher at the Institute of Robotics and Industrial Informatics (IRI), CSIC-UPC. He is the director of the Automatic Control Department and the head of the Research Group on Advanced Control Systems (SAC) at

UPC. He has made an important scientific contributions in the areas of fault diagnosis and fault tolerant control, using interval and linear-parameter-varying models exploiting setbased approaches. He is currently the chair of the IFAC SafeProcess TC Committee 6.4. He was the general chair of the 3rd IEEE Conference on Control and Fault-Tolerant Systems (SysTol 2016) and the IPC chair of IFAC SafeProcess 2018.



Radouane Khemmar graduated from the University of Poitiers in 2002 with an engineering degree in electronics, a post-graduate diploma (DEA) in industrial computing and a European Master's degree in image processing. He completed a PhD in image processing and computer vision at the University of Strasbourg in 2005. He then obtained a Habilitation à Diriger des Recherches (HDR) in computer vision, artificial intelligence and smart mobility from the Université Rouen Normandie in 2022. He worked

as instructor at CIES Alsace from 2002 to 2005. From 2005 to 2007, he held the position of Temporary Teaching and Research Associate (ATER) for 2 years at the University of Strasbourg, as well as a PostDoc in electronic document management (EDM) and dematerialization with the Jouve Group from 2007 to 2008. Subsequently, he held several positions in industry as a project manager with major industrial accounts (JOUVE, THALES, ALTEN). Since 2009, he has been an associate professor at ESIGELEC, specializing in embedded systems, mobile robotics and smart mobility. His research focuses on computer vision, mobile robotics, artificial intelligence and smart mobility.



Rihab El Houda Thabet obtained the engineering degree in 2010 and the M.Sc. degree in automatic and intelligent techniques in 2011 from Gabs National Engineering School (Tunis). She received the Ph.D. degree in Automatic, Production, Signal and Image, Cognitive Engineering from the University of Bordeaux in 2014. She was a Research Engineer at IRSEEM, Rouen from 2015 to 2016 and from 2018 to 2020.

Since 2021, she joined ESIGELEC as an associate professor. Her research deals with set membership techniques for the monitoring of uncertain systems.



Anh-Tu Nguyen received his PhD degree in Automatic Control in 2013 from the University of Valenciennes, France. From February 2014 to August 2018, he was a postdoctoral researcher at the CNRS laboratories LAMIH UMR 8201, Valenciennes, France and LS2N UMR 6004, Nantes, France. Since September 2018, he has been an Associate Professor at INSA Hauts-de-France, Université Polytechnique des Hauts-de-France, and a researcher at the LAMIH laboratory, Valenciennes, France. Dr. Nguyen

was a visiting researcher at various international institutions: European Center for Soft Computing, Spain (May-July 2014); University of Electro-Communications, Tokyo, Japan (October-December 2014); Tokyo Institute of Technology, Tokyo, Japan (November-December 2015); Beihang University, Beijing, China (July 2017, October 2019); Shanghai Maritime University, China (August 2017); The Hong Kong Polytechnic University (August-December 2022). Dr. Nguyen is a Senior Member of the IEEE. He has served as an Associate Editor for the IEEE Transactions on Intelligent Transportation Systems, the IFAC journal Control Engineering Practice, the IET Journal of Engineering, the SAE International Journal of Vehicle Dynamics, Stability, and NVH, the Springer Automotive Innovation, and a Guest Editor for special issues in various international journals. Dr. Nguyen's research interests include robust control and estimation, cybernetics control systems, human-machine shared control with a strong emphasis on mechatronics applications.



Madiha Nadri-Wolf received the M.S. degree in automatic control from Université Claude Bernard Lyon 1, Villeurbanne, France, in 1997. In 2001, she received a Ph.D. degree from the same university. After working one year as a Lecturer, she joined the Division of Applied Mathematics and Process Control at IFP Energies Nouvelles as a Research Assistant. Since 2005, she is an Assistant Professor in automatic control at the Department of Electrical and Chemical Engineering, at Université Claude Bernard Lyon 1.

Her research interests include state estimation problems, nonlinear observers, identification and Lyapunov stability of nonlinear systems, especially in biological systems and automotive application.

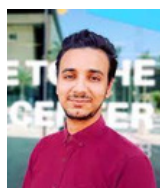


Petrit Rama is a PhD Researcher in Machine Learning & Autonomous Driving at the University of Kaiserslautern. He holds a Bachelor's and Master's degree in Computer Engineering from the University of Prishtina (Kosova). Since November 2018, he has been working as a research associate at RPTU Kaiserslautern-Landau. His research interests encompass decision-making systems for autonomous driving, maneuver prediction using Graph Neural Networks, graph representation learning, development of a Digital Twin for autonomous and

connected driving, ontology-based information modeling of a Digital Twin, and the development of AI techniques and algorithms for knowledge extraction and new inference rules in data semantics.



Lara Jaber received the telecommunications and computer engineering degree in 2021 from the Lebanese university in Lebanon. During the same year she received her master 2 Signal and image processing in collaboration between Centrale Supélec and Paris Saclay. She is a PhD candidate at Université-D'Evry Val d'Essonne-Paris-Saclay at the IBISC laboratory. She is interested in the development of intelligent observers in the framework of her current Phd :Hybrid approaches model-data for the fault-tolerant control of autonomous vehicles. Her work is integrated within the ANR ArtISMo Project supervised by D. Ichalal, L. Nouvelière and N. Ait Oufroukh.



Hichem Bessafa holds an engineering degree in automatics from the National Polytechnic School of Algiers and a Master's degree in ATSI (Automatique, Traitement du Signal et des Images) from the University of Paris Saclay. Currently, he is pursuing his research as a doctoral candidate at the University of Lorraine, within the Research Center in Automation of Nancy (CRAN). His research interests lie in the development of estimation algorithms for autonomous

and connected vehicles, using hybrid methods that combine classical observers and machine learning in the context of the ANR ArtISMo project, under the supervision of Ali Zemouche and Cédric Delattre, with the participation of Zehor Belkhatir.

Second Workshop on Intelligent Estimation Algorithms for Smart Mobility ArtISMo

State Estimation and Control of Autonomous Connected Vehicles: Advances and Challenges

PROGRAM & PRESENTATIONS DETAILS

20 OCTOBER 2023

8:15	Coffee break
8:45	Speaker Rihab El Houda Thabet "Filtered High Gain Interval Observer for LPV Systems with Bounded Uncertainties "
9:30	Speaker Anh-Tu Nguyen " Human-in-the-Loop Driving Control Design for Intelligent Vehicles Under Multiple System Constraints"
10:15	Speaker Petrit Rama "MALE-A: Stimuli and Cause Prediction for Maneuver Planning via Graph Neural Networks in Autonomous Driving"
11:00	Coffee break-Poster session
11:45	Speaker Madiha Nadri-Wolf "How Control Theory Learned to Stop Worrying and Succumbed to AI?"
13:00	Lunch break
14:30	Speaker Vicenç Puig (Online) "Safe Coordination of Autonomous Vehicles"
15:15	Visit of the IUT departments
16.30	ArtISMo Review Meeting
17.30	Workshop closing

Second Workshop on Intelligent Estimation Algorithms for Smart Mobility ArtISMo

State Estimation and Control of Autonomous Connected Vehicles: Advances and Challenges

PROGRAM & PRESENTATIONS DETAILS

19 OCTOBER 2023

11:30	Welcome reception
13:00	Lunch break
14:30	Opening speech
14:45	Speaker Lara Jaber <i>"A New High Gain Observer Based on Artificial Delays"</i>
15:30	Speaker Hichem Bessafa <i>"Estimation Algorithms applied to Autonomous Connected Vehicles"</i>
16:15	Coffee break
16:45	Speaker Rajesh Rajamani (Online) <i>"Challenging Estimation Problems in Vehicle Autonomy"</i>
17:30	Speaker Redouane Khemmar <i>"AI-based Environnement Perception for Autonomous Vehicle Parameters Estimation"</i>
18:15	Technical discussions

Intelligent Estimation Algorithms for Smart Mobility - ArtISMo Project

Context of the project: The development of controllers with high performance and reliability for autonomous and connected vehicles will require real-time measurements or estimates of many variables on each vehicle. Examples of variables that are needed for feedback include longitudinal distances, velocities, and accelerations of other nearby vehicles; lateral position of the vehicle in its own lane; vehicle yaw angle; slip angle; yaw rate; steering angle; lateral acceleration; and roll angle. There are also environmental variables which need to be measured such as tire-road friction coefficient, snow cover on road, and the presence of unexpected obstacles. Measurement of all of the above variables requires significant expense. Indeed, some of the sensors above, such as slip angle and roll angle, can be extremely expensive to measure, requiring sensors that cost thousands of dollars. For example, the Datron optical sensor for measurement of slip angle has a price over 10ke. In addition, several variables cannot be measured due to the unavailability of sensors (at any cost). Examples include positions, and accelerations of cars that are further upstream (e.g. Lead car of a platoon). Only the position of the immediately preceding car ahead can currently be measured. Furthermore, autonomous and connected vehicle requires highly reliable sensors and actuators. Failure of any one sensor or actuator, due to faults, cyber-attacks or denial of service, can cause a disastrous accident. Hence reliable fault diagnostic and fault handling systems are also needed. Such systems cannot be based on hardware redundancy which requires many extra copies of the same sensors. Instead, they need to rely on estimation algorithms and analytical redundancy. For all the above reasons, the development of intelligent estimation algorithms is highly important for autonomous vehicles.

PROJECT COORDINATOR

Ali ZEMOUCHE - Centre de recherche en automatique de Nancy (CRAN)

PARTNERS :

FAAR SAS

CRAN - Centre de Recherche en Automatique de Nancy

IBISC - Informatique, BioInformatique, Systèmes Complexes

IRSEEM - Institut de recherche en systèmes électroniques embarqués

Project ANR 20-CE48-0015

Grant : 514,283 euros

Beginning and duration of the project: February 2021 - 48 Months

